

Year 2 Assessed Problems

Semester 1

Assessed Problems 8

SOLUTIONS TO BE SUBMITTED
ON CANVAS BY

Tuesday 3rd December
at 17:00

4 Line integrals

★Problem 4.1 Line integrals

- (i) Evaluate the line integral

$$\int_C \mathbf{F} \cdot d\mathbf{r} ,$$

where C is the helix given by

$$\mathbf{r}(\theta) = a \cos \theta \hat{\mathbf{x}} + a \sin \theta \hat{\mathbf{y}} + a\theta \hat{\mathbf{z}} \quad (0 \leq \theta \leq 2\pi)$$

whilst $\mathbf{F}(\mathbf{r}) = \mathbf{r}$.

[3]

- (ii) Evaluate the line integral from the point $(0, 0, 0)$ to the point $(1, 1, 1)$ of the function $\mathbf{F}(\mathbf{r}) = x^2 \hat{\mathbf{x}} + y^2 \hat{\mathbf{y}} + z^2 \hat{\mathbf{z}}$ along the following curves C :

(a) The straight line $\mathbf{r}(u) = u\hat{\mathbf{x}} + u\hat{\mathbf{y}} + u\hat{\mathbf{z}}$, where $0 \leq u \leq 1$.

[3]

(b) The parabola $\mathbf{r}(u) = u\hat{\mathbf{x}} + u\hat{\mathbf{y}} + u^2\hat{\mathbf{z}}$, where $0 \leq u \leq 1$.

[4]

3 Assessed – Statistical physics

Problem 3.1 Magnetic atoms in an external field

Consider a system consisting of 5 *distinguishable* molecules (that could mean that each of them is pinned to a specific location). Each molecule has a spin $1/2$ and a permanent magnetic moment μ . The magnetic moment can therefore take only two orientations: $\uparrow$ or $\downarrow$. In the presence of a (vertical) external magnetic field $\mathbf{H}$, the magnetic energy of a molecule is $-\mu \cdot \mathbf{B}$, where $\mathbf{B} = \mu_0 \mathbf{H}$, so that the energy for the orientation $\uparrow$ of the magnetic moment is $-\mu B$. We neglect other types of energies such as the kinetic and interaction energies (either interaction between the magnetic dipoles or by other means), and the molecules are isolated from their surroundings.

- (i) Give an expression for and calculate the statistical weight of the macrostate that has fixed total magnetic energy $U_{\text{tot}} = -\mu B$. [2]
- (ii) We replace the first molecule with a molecule having a magnetic moment of magnitude $\mu' = 3\mu$ (but still with spin $1/2$), which maybe denoted $\uparrow\uparrow$ or $\downarrow\downarrow$. The new system consists of one molecule with magnetic moment 3μ and four molecules each with moment μ . Calculate the statistical weight of the macrostate that has fixed total magnetic energy $U_{\text{tot}} = -\mu B$. You will either give an expression for it or list all the accessible microstate(s). [2]
- (iii) Similar to part 2, but the new molecule has a magnetic moment $\mu' = 5\mu$, denoting this as $\uparrow\uparrow\uparrow$ or $\downarrow\downarrow\downarrow$. List the accessible microstate(s). [1]

Now, consider two systems made of distinguishable molecules. System A has five molecules with one of them having magnetic moment $\mu' = 5\mu$ and the remaining having μ each; system B has five molecules each with magnetic moment μ . Initially, the systems are prepared to have the total energy $-\mu B$ each. We then bring the two systems, which are otherwise isolated, into thermal contact so that energy can be exchanged, but not molecules.

- (iv) Calculate the number of microstates for the *thermalised* system $A + B$. [1]
- (v) Find in which direction heat flows between A and B during thermalisation. [4]
Hint: you can find the possible configurations of A (or B) and calculate the mean energy of A (or B) after thermalisation.